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五种商品意大利面在碱水煮制下的尺寸变化、时间进程与中式面条适配性：一项探索性厨房比较

Dimensional Changes, Time Course, and Chinese-Noodle Applications of Five Commercial Pasta Types Cooked in Alkaline Water: An Exploratory Kitchen Comparison

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摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房比较在统一碱水条件下考察五种商品意大利面的尺寸变化、建议煮制时间和中式面条应用潜力。Angel Hair、Thin Spaghetti、Spaghetti、Linguine和Fettuccine均在1 L沸水加入4 g小苏打的条件下煮制。所报告的生面/熟面尺寸分别为1.2/1.8、1.4/2.1、1.8/2.8、3.1/4.6和4.6/6.8 mm；对应时间为5、9、15、17和18分钟。平均尺寸膨胀倍数为1.50。Angel Hair在3至8分钟内由1.5增加至1.9 mm，但5分钟被选为感官端点，说明最大膨胀并不等同于最佳口感。名义尺寸与建议时间呈描述性正相关（Pearson r = 0.82），但样本仅五种且混合了圆面直径与扁面宽度，因此结果只应作为实践参考。

Abstract: This exploratory kitchen comparison examined dimensional change, recommended cooking time, and potential Chinese-noodle applications for five commercial pasta formats under a standardized alkaline condition. Angel Hair, Thin Spaghetti, Spaghetti, Linguine, and Fettuccine were cooked in 1 L boiling water containing 4 g baking soda. Reported dry/cooked dimensions were 1.2/1.8, 1.4/2.1, 1.8/2.8, 3.1/4.6, and 4.6/6.8 mm, with cooking times of 5, 9, 15, 17, and 18 minutes. Mean dimensional expansion was 1.50. Angel Hair increased from 1.5 to 1.9 mm from 3 to 8 minutes, although 5 minutes was selected as the sensory endpoint, indicating that maximum swelling did not equal optimal texture. Nominal size and recommended time showed a descriptive positive association (Pearson r = 0.82), but the five-item dataset mixed diameter and width measurements and should be treated as a practical reference only.

Keywords: alkaline cooking; caliper measurement; dimensional expansion; pasta format

1. Background / 背景

Commercial pasta formats differ in thickness, geometry, and conventional cooking instructions. The study compared five formats under a common alkaline condition to examine time, dimensional change, and possible Chinese-noodle applications.

2. Materials and Methods / 材料与方法

Products were independently purchased and cooked in 1 L boiling water with 4 g baking soda. Time was recorded at the author-defined al dente endpoint. Diameter was recorded for round pasta and width for Linguine and Fettuccine. Strand count, replication, calibration, and caliper pressure were not documented.



Figure 2. Compared pasta products.

3. Results / 结果

Type	Dry	Cooked	Ratio	Min
Angel Hair	1.2	1.8	1.50x	5
Thin Spag.	1.4	2.1	1.50x	9
Spaghetti	1.8	2.8	1.56x	15
Linguine*	3.1	4.6	1.48x	17
Fettuccine*	4.6	6.8	1.48x	18

* Flat-pasta values are widths; round-pasta values are diameters.

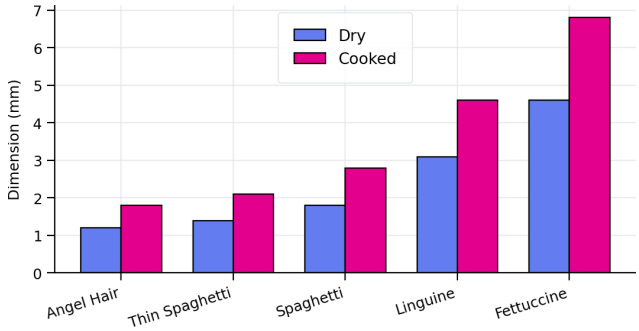


Figure 3. Dry and cooked dimensions.

Expansion ratios clustered from 1.48 to 1.56. Recommended time increased from 5 to 18 minutes. The descriptive correlation between nominal size and recommended time was r = 0.82, but the dataset is too small and heterogeneous for inferential claims.

4. Angel Hair Time Course / 时间进程

Angel Hair readings at 3, 4, 5, 6, 7, and 8 minutes were 1.5, 1.6, 1.7, 1.8, 1.8, and 1.9 mm. A firm center remained at 3 minutes, and 5 minutes was selected as the recommended sensory endpoint.

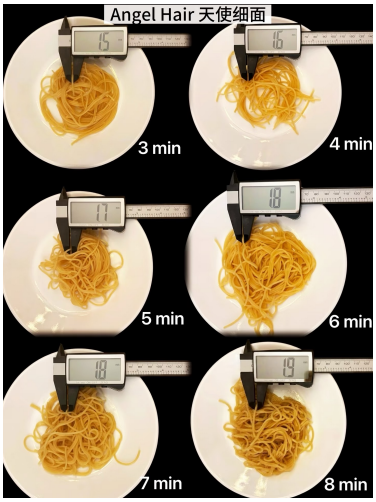


Figure 4. Angel Hair measurements.

5. Culinary Applications / 应用建议

Angel Hair and Thin Spaghetti suited fine-noodle applications; standard Spaghetti was the most versatile; Linguine and Fettuccine suited mixed, stir-fried, or broad-noodle styles. These culinary suggestions do not imply equivalence in ingredients or structure.



Figure 5. Original applications.

6. Discussion / 讨论

A memorable approximate 1.5-fold dimensional change emerged, but width and diameter are not equivalent physical measures and cannot support conclusions about cross-sectional area, volume, or water uptake.

7. Limitations / 局限性

Strand count, replication, measurement location, caliper pressure, and calibration were not documented. One author selected the sensory endpoint, and product formulation could influence results.

8. Conclusion / 结论

Under 4 g/L baking soda, recommended times ranged from 5 to 18 minutes and mean dimensional expansion was approximately 1.50. Future work should use repeated measurements, controlled caliper pressure, cooking-loss analysis, and instrumental texture testing.

Publication Information / 出版信息

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